Derivability from a primitive recursive axiom set is primitive recursive

The Open Logic Project contributors

openlogic:incompleteness/arithmetization-syntax:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author proposition labelled prop:prf-prim-rec (1679-1684); the proof span 1686-1700 is the author proof environment that immediately follows it.

Editorial note. External reference numbers render the unit-internal labels with hyphens in place of colons (pipeline number charset); the Source reference lines below the body give each label's exact source span.

Editorial scope. One original result of the Open Logic Project unit incompleteness/arithmetization-syntax, counted once: derivability from a primitive recursive axiom set is primitive recursive. Only the counted proposition and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Derivability predicate. The property Deriv(x), holding when x is the Godel number of a correct natural deduction derivation, is primitive recursive; the unit proves this as its own proposition.. It is the first conjunct of the displayed definition of Prf_Gamma (1689-1690).
- End formula and open assumptions. EndFmla(x) returns the Godel number of the end formula of the derivation coded by x, and OpenAssum(z, x) says z codes an undischarged assumption of that derivation; both are primitive recursive.. They tie the derivation to the sentence y and make the side condition a bounded universal over $z < x$ (1689-1699).
- Primitive recursive axiom sets. Membership in Gamma is decided by a primitive recursive predicate R_Gamma.. The bounded universal requires every open assumption z to satisfy R_Gamma (1686-1688, 1691-1699).

Source reference. Original proposition prop-deriv: sources/olp-incompleteness-arithmetization-syntax/pinned-original-source.tex, source lines 1324–1324. The author proof cites the unit's proposition that the derivability predicate is primitive recursive. The load-bearing work of that predicate is carried by the separately proved proposition of the same unit; the counted proof retains its own assembly in full.

Proposition 0.1. *Suppose* Γ *is a primitive recursive set of !!sentences. Then the relation* $\mathrm{Prf}[\Gamma](x, y)$ *expressing "*x *is the code of !!a derivation* δ *of* $!A$ *from !!undischarged assumptions in* Γ *and* y *is the Gödel number of* $!A$*" is primitive recursive.*

Proof. Suppose "$y \in \Gamma$" is given by the primitive recursive predicate $R_\Gamma(y)$. We have to show that $\mathrm{Prf}[\Gamma](x, y)$ which holds iff y is the Gödel number of a sentence $!A$ and x is the code of a natural deduction !!derivation with end !!formula $!A$ and all !!undischarged assumptions in Γ is primitive recursive.

By prop-deriv, the property $\mathrm{fn}\, Deriv(x)$ which holds iff x is the Gödel number of a correct !!derivation δ in natural deduction is primitive recursive. Thus we can define $\mathrm{Prf}[\Gamma](x, y)$ by

$$\begin{aligned}\mathrm{Prf}[\Gamma](x, y) \leftrightarrow\ & \mathrm{fn}\, Deriv(x) \wedge \mathrm{fn}\, EndFmla(x) = y \wedge \\ & \forall z < x(\mathrm{fn}\, OpenAssum(z, x) \rightarrow R_\Gamma(z)).\end{aligned}$$

□

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